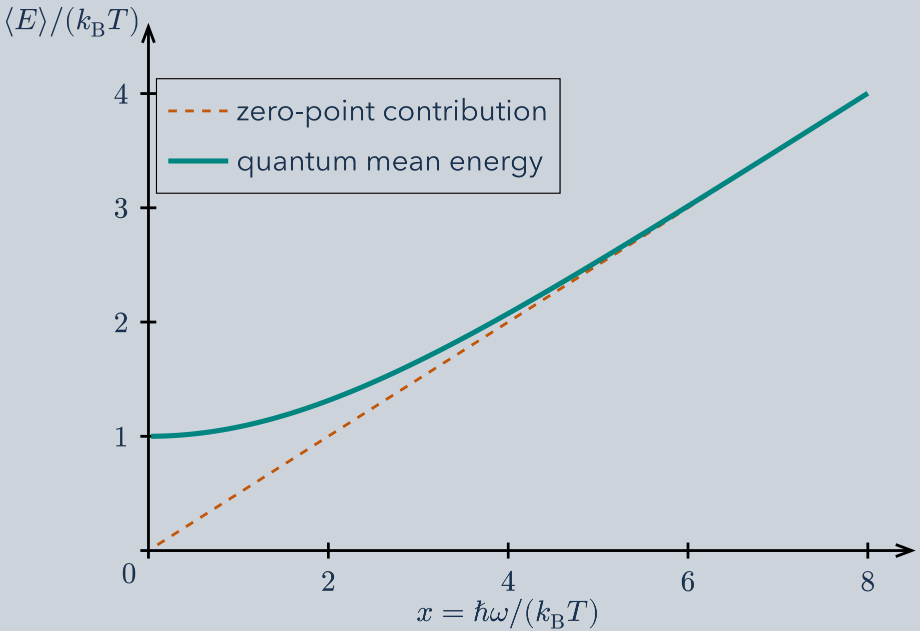
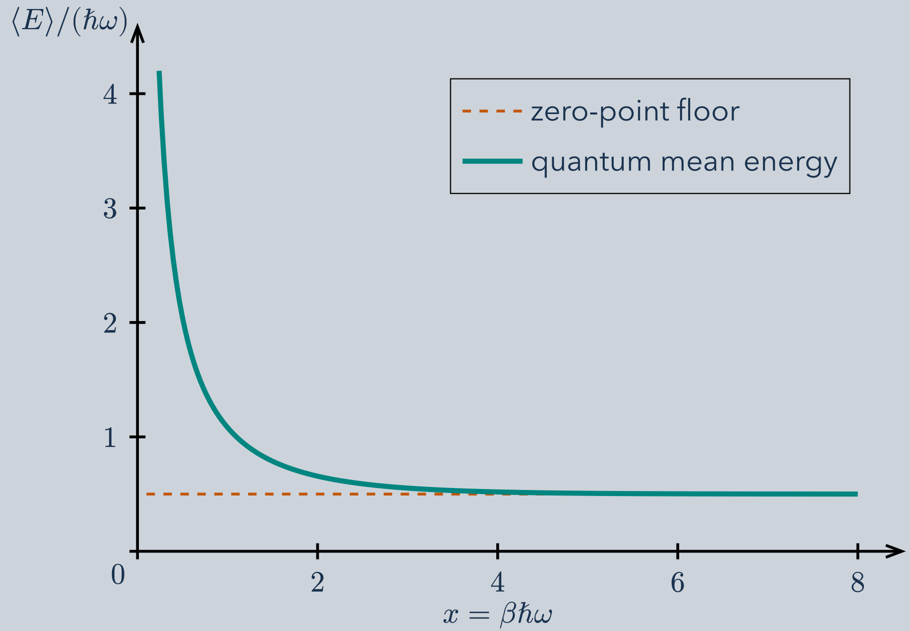
One energy formula connects thermal motion with zero-point motion. Compare the level spacing $\hbar\omega$ with the thermal energy $k_\mathrm{B}T$.

1 Increase frequency at fixed temperature



At small level spacing, $\langle E\rangle \approx k_\mathrm{B}T$: classical equipartition. At large spacing the zero-point term dominates. The energy rises monotonically.

2 Cool at fixed frequency



Move right by increasing $\beta = 1/(k_\mathrm{B}T)$. Thermal excitations freeze out; the mean energy approaches $\hbar\omega/2$, rather than zero.

$\langle E\rangle = \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega}-1}\right)$. **The same horizontal variable, two energy units.** ω is angular frequency; $\hbar$ is the reduced Planck constant.